

SocietyXAI: Causal Explainability of Belief Dynamics and Emergent Consensus in Multi-Agent Generative Systems

1. Core Idea

SocietyXAI is a research framework for understanding how societies of generative AI agents form, change, and converge on collective decisions.

Instead of treating a multi-agent system as a black box that simply produces a final consensus, SocietyXAI investigates the internal dynamics of the deliberation:

- Why did an agent change its belief?
- Which agent influenced another agent?
- Which argument or evidence item caused a belief transition?
- Did agents converge because of valid evidence or because of social pressure?
- Did one dominant agent disproportionately control the final decision?
- Did an initially incorrect majority pressure a correct minority agent into changing its answer?
- Can a weaker but persuasive agent influence a stronger reasoning model?
- Does model diversity make an artificial society more robust?
- Can an adversarial agent trigger a cascade that eventually changes the entire group's decision?

The central idea is to transform multi-agent deliberation from something that is merely observable into something that is causally explainable.

2. Problem Statement

Multi-agent Large Language Model systems increasingly use multiple AI agents to collaboratively reason, debate, critique, negotiate, and reach collective decisions.

Existing systems can observe:

- messages exchanged between agents,
- voting patterns,
- stance changes,
- confidence changes,
- final consensus,
- task accuracy.

However, observing these changes does not establish why they occurred.

For example, if five AI agents initially disagree but eventually reach the same conclusion, several fundamentally different mechanisms could have produced that consensus:

1. Agents independently evaluated strong evidence and rationally converged.
2. Agents followed the majority.
3. A highly confident agent anchored the discussion.
4. A powerful model persuaded weaker models.
5. Agent ordering influenced later participants.
6. A specific piece of evidence caused several belief changes.
7. A biased or adversarial agent initiated an influence cascade.
8. Agents developed an echo chamber and repeatedly reinforced the same claim.
9. The perceived authority or role of an agent influenced other agents independently of the quality of its argument.

Therefore, agreement alone cannot demonstrate that a multi-agent reasoning process was sound.

SocietyXAI addresses this problem by modeling multi-agent deliberation as an evolving causal belief network and performing controlled interventions on the system to determine what actually caused downstream belief changes and consensus.

3. Central Research Question

«When a society of generative AI agents reaches consensus, can we causally explain why individual agents changed their beliefs, who influenced whom, whether evidence or social dynamics drove convergence, and why the society ultimately became correct or collectively wrong?»

This produces several secondary research questions.

RQ1: Causal Influence

Which agents, messages, arguments, and evidence items have the greatest causal influence on other agents?

RQ2: Evidence vs Social Influence

Can SocietyXAI distinguish evidence-driven convergence from convergence caused by conformity, persuasion, anchoring, or authority effects?

RQ3: Model Capability

Are stronger reasoning models less susceptible to influence from weaker models?

Alternatively, can weaker but highly confident or persuasive agents influence stronger models?

RQ4: Artificial Groupthink

Under what conditions do multi-agent systems produce false consensus, where most or all agents agree on an incorrect conclusion?

RQ5: Minority Correctness

Can a single correct minority agent convince an incorrect majority?

Conversely, can an incorrect majority cause the correct minority agent to abandon its answer?

RQ6: Model Diversity

Are heterogeneous societies containing different LLM families more robust than societies containing multiple instances of the same model?

RQ7: Architecture

How do different multi-agent communication architectures affect influence, conformity, consensus quality, and error propagation?

RQ8: Adversarial Influence

Can one adversarial or biased agent manipulate the collective decision of otherwise reliable agents?

4. Main Research Contribution

SocietyXAI does not merely observe conversations and construct an influence graph afterward.

It performs controlled counterfactual interventions.

Suppose:

Agent A $\rightarrow$ Agent B $\rightarrow$ Agent C $\rightarrow$ Consensus X

SocietyXAI can intervene by removing Agent A's contribution and replaying the deliberation:

do(Agent A message = removed)

If the resulting trajectory becomes:

Agent B $\rightarrow$ Agent C $\rightarrow$ Consensus Y

then Agent A's contribution has measurable causal importance.

This allows SocietyXAI to move beyond:

"Agent B changed after Agent A spoke."

toward:

"Removing Agent A's argument prevented Agent B's belief transition and changed the final consensus."

The second statement provides substantially stronger evidence of influence.

5. Belief Dynamics

Every agent maintains an explicit state during deliberation.

Conceptually:

Agent State at Time t

Role

Model

Current position

Confidence

Evidence used

Arguments received

Current reasoning

Previous position

Previous confidence

For an agent A_i at round t, SocietyXAI records:

$B_i(t) = \{$
 position,
 confidence,

```
evidence,  
reasoning state  
}
```

This allows the system to study belief trajectories, rather than examining only final votes.

Example:

ROUND 0

Agent A	YES	0.91
Agent B	NO	0.82
Agent C	NO	0.67
Agent D	YES	0.61
Agent E	NO	0.73

After Agent A presents an argument:

ROUND 1

Agent A	YES	0.93	
Agent B	NO	0.76	
Agent C	YES	0.58	← belief flip
Agent D	YES	0.79	← confidence increase
Agent E	NO	0.65	

SocietyXAI then investigates what caused those changes.

6. Causal Intervention Engine

The central technical component of SocietyXAI is an intervention and counterfactual replay engine.

Several forms of intervention can be studied.

6.1 Agent Intervention

Remove an entire agent.

Example:

Original society:

$A + B + C + D + E \rightarrow \text{Consensus X}$

Counterfactual:

$B + C + D + E \rightarrow \text{Consensus Y}$

This measures the causal importance of Agent A to the collective outcome.

6.2 Message Intervention

Remove one specific argument generated by an agent.

Agent A message #14 $\rightarrow$ removed

The deliberation is replayed from that point.

If downstream beliefs change substantially, the message was causally influential.

6.3 Evidence Intervention

Keep the agent but remove or replace the evidence supporting its argument.

This allows us to distinguish:

influence of the agent

from

influence of the evidence.

6.4 Confidence Intervention

Hide explicit confidence information.

For example:

"I believe option A."

versus

"I am 97% confident that option A is correct."

If other agents behave differently, confidence signalling itself may influence collective reasoning.

6.5 Ordering Intervention

Change who speaks first.

Example:

Original:

$A \rightarrow B \rightarrow C \rightarrow D$

Counterfactual:

$D \rightarrow C \rightarrow B \rightarrow A$

This allows SocietyXAI to detect anchoring and first-speaker effects.

6.6 Identity / Role Intervention

Keep an argument identical while changing the apparent source.

For example:

"This evidence supports X."

Source: Expert Agent

versus:

"This evidence supports X."

Source: Junior Analyst Agent

If identical arguments cause different belief changes depending on the attributed speaker, this provides evidence of an authority or role bias.

6.7 Majority Visibility Intervention

Agents can either:

- see other agents' current votes,
- see only their arguments,
- or reason independently.

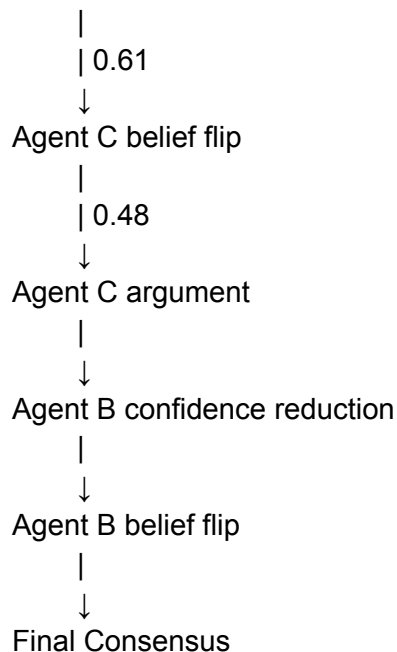
Comparing these conditions allows SocietyXAI to estimate the effect of majority information on conformity.

7. Causal Belief-Propagation Graph

SocietyXAI generates a dynamic causal influence graph.

Example:

Agent A / Message 1



This allows us to identify influence cascades.

A single influential contribution may indirectly alter the entire society.

8. Social Phenomena SocietyXAI Can Detect

Evidence-Driven Convergence

Agents independently update their beliefs because new evidence supports a conclusion.

This represents the desirable form of consensus.

Conformity

An agent abandons its independently supported position primarily because the majority disagrees.

Anchoring

Early arguments disproportionately influence later reasoning.

Agent Dominance

One agent has disproportionately high causal influence over collective decisions.

Authority Bias

Agents respond differently to identical information depending on which role or model supposedly produced it.

Echo Chambers

Agents repeatedly reinforce existing arguments without meaningful introduction of new evidence.

Belief Cascades

Example:

A convinces B
↓
B convinces C
↓
B + C create majority
↓
D changes
↓
E changes
↓
Consensus

False Consensus

All or most agents agree, but the final answer is incorrect.

The research question then becomes:

«How did an artificial society collectively reason itself into the wrong answer?»

Adversarial Manipulation

A deliberately biased or adversarial agent introduces misleading reasoning or evidence and attempts to influence the collective decision.

SocietyXAI measures how far this influence propagates.

9. Heterogeneous AI Societies

One of the major proposed experiments is to avoid using only multiple copies of the same LLM.

SocietyXAI can construct heterogeneous artificial societies containing different model families and capability levels.

Conceptually:

Agent A → Frontier / strong reasoning model

Agent B → Different frontier model
Agent C → Medium-capability model
Agent D → Small/open-weight model
Agent E → Domain-oriented model

Possible providers and model families can include, depending on availability during experimentation:

- OpenAI models
- Gemini models
- Claude models
- open-weight Llama-family models
- Qwen-family models
- Mistral-family models
- other reproducible open models

The exact model IDs and versions should be frozen and documented when experiments begin.

10. Strong and Weak Agents

Agents should not arbitrarily be labelled "strong" or "weak."

Model capability should be operationally defined using:

1. publicly available benchmark results,
2. comparable official evaluations where available,
3. independent benchmark results where appropriate,
4. controlled pre-evaluation performed as part of SocietyXAI.

Possible capability dimensions include:

- logical reasoning,
- factual reasoning,
- mathematical reasoning,
- instruction following,
- domain knowledge,
- resistance to misleading information,
- calibration,
- long-context reasoning.

Therefore an agent might be:

Agent A

Model: Model X

Role: Evidence Analyst

Reasoning capability: High

Domain capability: Medium

Initial confidence: 0.81

while another could be:

Agent B

Model: Model Y

Role: Risk Critic

Reasoning capability: Medium

Domain capability: High

Initial confidence: 0.64

This produces artificial societies with real capability diversity, rather than simply prompting identical models to pretend to be different personalities.

11. Capability vs Social Behaviour

An important distinction should be maintained between:

Intrinsic model capability

and

assigned social behaviour.

A strong reasoning model could deliberately be assigned a cautious or minority role.

A weaker model could be given a highly assertive communication style.

This enables experiments such as:

«Can a weak but highly assertive agent convince stronger agents to accept an incorrect conclusion?»

And:

«Can a highly capable but cautious minority agent successfully correct an incorrect majority?»

12. Homogeneous vs Heterogeneous Societies

Two major experimental conditions can be compared.

Homogeneous Society

Model A

Model A

Model A

Model A

Model A

Heterogeneous Society

Model A

Model B

Model C

Model D

Open Model E

We can then investigate whether diversity affects:

- consensus accuracy,
- false-consensus rate,
- conformity,
- influence concentration,
- adversarial robustness,
- belief diversity,
- convergence speed,
- confidence calibration.

A particularly interesting research question is:

«Does model diversity make artificial societies more epistemically robust, or does it merely make consensus dynamics more complex and unpredictable?»

13. Four Existing Multi-Agent Case Studies

SocietyXAI can be evaluated across four previously developed multi-agent environments.

These environments are used as case studies, not as the central contribution of the paper.

The central research subject remains causal explainability of multi-agent decision-making.

Case Study 1: Healthcare / Hospital Decision Society

Possible roles:

General Physician

Specialist

Evidence Agent

Risk Agent

Treatment Planning Agent

Architecture:

Specialist consultation model

Agents can initially reason independently before being exposed to other opinions.

This environment can test:

- expert disagreement,
- minority opinions,
- confidence effects,
- evidence-driven convergence,
- specialist dominance.

14. Case Study 2: Financial Decision Society

Possible roles:

Growth Analyst

Risk Analyst

Portfolio Agent

Macroeconomic Analyst

Compliance Agent

Architecture:

Committee / risk-review model

Agents sequentially critique a proposed decision.

This environment can test:

- risk versus reward disagreements,
- sequential influence,
- analyst dominance,
- confidence effects,
- adversarial information.

15. Case Study 3: Court / Legal Reasoning Society

Possible roles:

Prosecution Agent
Defence Agent
Evidence Analyst
Precedent Researcher
Judge / Adjudicator

Architecture:

Adversarial deliberation model

This environment is particularly useful because disagreement is intentionally built into the architecture.

SocietyXAI can examine:

- adversarial reasoning,
- evidence influence,
- argument strength,
- judge susceptibility,
- role authority,
- belief shifts after rebuttals.

16. Case Study 4: ESG / Sustainability Management Society

Possible roles:

Sustainability Agent
Financial Agent
Operations Agent
Legal / Compliance Agent
Community Impact Agent

Architecture:

Stakeholder negotiation model

Agents have deliberately conflicting objectives.

For example, a decision may involve:

- environmental impact,
- operating costs,
- profitability,
- regulatory compliance,
- community impact,
- long-term sustainability.

SocietyXAI can determine whether the final consensus represents balanced evidence-based reasoning or whether one stakeholder dominates the decision.

17. Why Four Domains Matter

The four case studies should not simply represent four datasets.

They can represent four different interaction architectures:

Hospital

→ specialist consultation

Finance

→ committee / sequential risk review

Court

→ adversarial debate

ESG

→ stakeholder negotiation

This allows SocietyXAI to test whether causal influence phenomena generalize across fundamentally different multi-agent structures.

The experimental space therefore becomes:

Domain × Agent Architecture × Model Composition

This is substantially stronger than evaluating one fixed multi-agent debate architecture on multiple question datasets.

18. High-Value Experiments

Experiment A: Correct Minority vs Incorrect Majority

Agent A → CORRECT

Agent B → WRONG

Agent C → WRONG

Agent D → WRONG

Agent E → WRONG

Question:

Can the correct minority convince the society?

Or does majority pressure cause the correct agent to become wrong?

Experiment B: Incorrect Minority vs Correct Majority

Agent A → WRONG

Agent B → CORRECT

Agent C → CORRECT

Agent D → CORRECT

Agent E → CORRECT

Determine whether an influential incorrect agent can destabilize a correct society.

Experiment C: Strong Minority vs Weak Majority

One high-capability model disagrees with several weaker models.

Measure whether reasoning capability predicts causal influence.

Experiment D: Weak but Dominant Agent

A lower-capability model communicates with high confidence or assertiveness.

Determine whether social dominance can overpower reasoning capability.

Experiment E: Adversarial Agent Injection

Introduce one agent intentionally supplied with incorrect or manipulated evidence.

Measure:

- immediate influence,
- downstream propagation,
- belief cascades,
- final consensus,
- recovery capability.

Experiment F: Speaker Ordering

Run identical societies under different speaking orders.

Compare:

Strong model speaks first

vs

Strong model speaks last

This tests anchoring and ordering effects.

Experiment G: Confidence Visibility

Compare:

Arguments + confidence

vs

Arguments only

Determine whether explicit confidence signalling alters influence.

Experiment H: Identity Swap

Keep the argument identical but change the attributed agent identity or role.

This investigates whether agent authority affects reasoning independently of argument content.

Experiment I: Homogeneous vs Heterogeneous Society

Compare same-model societies against multi-model societies.

Measure robustness, accuracy, conformity and influence concentration.

Experiment J: False Consensus Autopsy

Identify experiments where every agent eventually converges on an incorrect answer.

SocietyXAI reconstructs the entire causal pathway responsible for the failure.

This can produce a causal autopsy of collective AI failure.

19. Proposed Metrics

SocietyXAI should evaluate more than final accuracy.

Potential metrics include:

Consensus Accuracy

Whether the collective decision is correct.

Belief Shift Rate

How frequently agents change their positions during deliberation.

Confidence Shift

Magnitude and direction of confidence changes.

Influence Concentration

Whether causal influence is distributed across agents or concentrated in a small number.

Conformity Index

Degree to which belief changes correlate causally with majority exposure rather than new evidence.

Evidence Alignment

Extent to which belief changes correspond to supporting evidence.

Counterfactual Agent Effect

Change in final outcome when an agent is removed.

Counterfactual Message Effect

Change in downstream beliefs when a specific message is removed.

Consensus Stability

Whether consensus survives controlled interventions.

Adversarial Robustness

Ability of the society to resist manipulated or incorrect agents.

Minority Recovery Rate

Probability that a correct minority can eventually move the group toward the correct conclusion.

False Consensus Rate

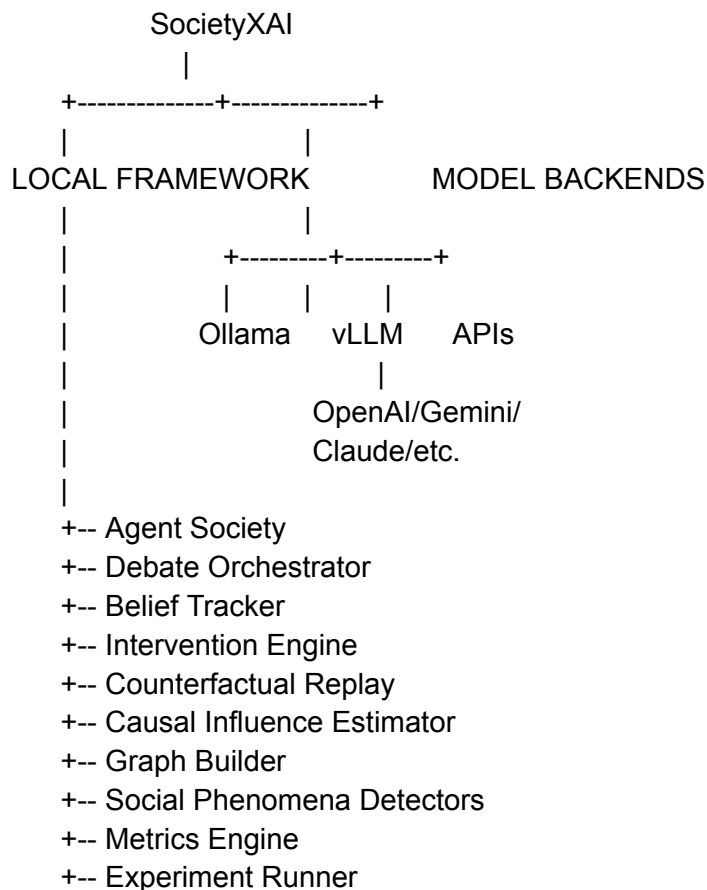
Frequency with which agents strongly agree on an incorrect conclusion.

The exact mathematical definitions of novel metrics must be formally established and validated rather than assigning arbitrary influence scores.

20. SocietyXAI Architecture

The framework should be local-first, model-agnostic, reproducible, API-optional and causality-centered.

Conceptually:



+-- Evaluation Suite

21. Local-First Does Not Mean Local-Models-Only

The SocietyXAI framework itself should run locally.

Potentially:

`pip install societyxai`

However, the framework should support interchangeable inference backends.

Possible modes:

SocietyXAI

|

- +-- Local Ollama models
- +-- Hugging Face models
- +-- local vLLM server
- +-- OpenAI-compatible endpoint
- +-- Gemini API
- +-- Claude API
- +-- other supported providers

Therefore:

Framework execution = local

Model inference = local or remote

This allows reproducibility without restricting experiments to smaller local models.

22. Proposed Package Structure

A future open-source implementation could be divided into:

`societyxai-core`

Responsible for:

- agent definitions,
- society construction,
- debate orchestration,
- belief tracking,
- interventions,
- causal analysis,
- experiment management,
- metrics.

societyxai-models

Model adapters for:

- local models,
- Ollama,
- Hugging Face,
- vLLM,
- proprietary APIs.

societyxai-lab

Interactive visualization and research dashboard.

23. Example Developer Interface

A possible future API could look like:

```
from societyxai import Society, Agent, Experiment
```

```
society = Society([
    Agent("prosecution"),
    Agent("defence"),
    Agent("evidence_analyst"),
    Agent("precedent_researcher"),
    Agent("judge")
])
```

```
experiment = Experiment(society)
```

```
result = experiment.run(case)
```

```
result.explain()
```

This is only a conceptual interface. The actual package design should follow after the research methodology is finalized.

24. SocietyXAI Lab

A visual research interface could display the evolving artificial society.

Example:

ROUND 0	ROUND 1	ROUND 2	ROUND 3
A	A	B	B
B	B	B	B
C	A	A	B
A	A	A	B

Selecting a belief transition could show:

Agent C

$A \rightarrow B$

Primary causal factor:

Agent 2, message #14

Counterfactual:

Removing message #14 prevented the transition in controlled replays.

Secondary factor:

Majority visibility

Estimated influence:

High

The interface could visualize:

- belief trajectories,
- causal graphs,
- confidence evolution,
- influence cascades,
- dominant agents,

- evidence propagation,
- consensus formation,
- counterfactual universes.

The dashboard would make SocietyXAI useful not only as a paper implementation but potentially as an audit and research tool for multi-agent systems.

25. API Usage and Experimental Cost

Large-scale counterfactual experimentation can require substantial inference because a single original debate may generate many alternative trajectories.

For example:

Original Debate

- |
- +-- remove Agent A
- +-- remove message M1
- +-- remove evidence E1
- +-- hide confidence
- +-- reorder speakers
- +-- swap identity
- +-- hide majority

Each intervention can require partial or complete replay.

Therefore paid APIs will likely be used for final experiments involving frontier models.

However, API cost can be controlled.

26. Development Strategy

During development:

Local / inexpensive models

↓

Build orchestration

↓

Debug belief tracking

↓

Develop intervention engine
↓
Validate causal graph generation
↓
Develop metrics
↓
Pilot experiments

There is no reason to spend premium API tokens while debugging the framework.

Once the methodology is frozen:

Final model selection
↓
Pilot API experiments
↓
Freeze parameters
↓
Official experimental runs
↓
Statistical evaluation

27. Counterfactual Caching

Counterfactual experiments should avoid regenerating unaffected history.

Suppose:

$R_0 \rightarrow R_1 \rightarrow R_2 \rightarrow R_3 \rightarrow R_4$
 |
 intervention

Rounds before the intervention can potentially be reused.

Only the affected trajectory needs regeneration:

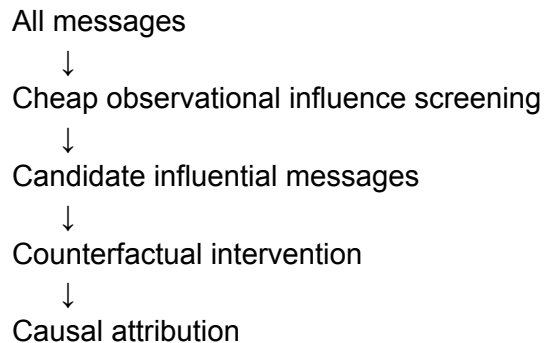
$R_0 \rightarrow R_1 \rightarrow R_2$
 \
 $R_3' \rightarrow R_4'$

This can significantly reduce API calls.

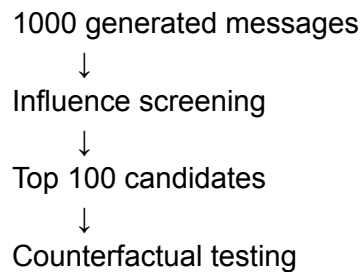
28. Adaptive Intervention Selection

It may be computationally infeasible to counterfactually test every generated message.

Therefore SocietyXAI can use a two-stage process:



For example:



The exact screening method would itself need careful design to avoid biasing causal analysis.

29. Local and Open Models as Scientific Assets

Open models are not merely a way to reduce costs.

They provide:

- reproducibility,
- exact model checkpoints,
- controllable inference parameters,
- repeatable experiments,
- local deployment,
- broader capability ranges.

SocietyXAI can therefore compare:

Small open model



Medium model



Large open model



Frontier proprietary model

One research question becomes:

«Does susceptibility to social influence decrease as model reasoning capability increases?»

A surprising alternative is also possible:

«More capable models may become more effective persuaders while remaining susceptible to sophisticated incorrect arguments.»

Either outcome would be scientifically interesting.

30. Experimental Reproducibility

For every experiment, SocietyXAI should record:

- exact model identifier,
- provider,
- access date,
- inference parameters,
- temperature,
- system prompt,
- agent role,
- communication topology,
- debate order,
- number of rounds,
- original evidence,
- intervention,
- resulting belief trajectory,
- final decision.

This is especially important for proprietary models because hosted models can change over time.

31. Expected Research Contributions

The project could potentially contribute:

1. A Causal Explainability Framework for Multi-Agent GenAI

A framework specifically designed to explain how collective decisions emerge.

2. Causal Belief-Propagation Graphs

A representation connecting agents, arguments, evidence and downstream belief transitions.

3. Counterfactual Multi-Agent Interventions

A methodology for estimating influence by replaying deliberations under controlled changes.

4. Artificial-Society Behaviour Analysis

Detection and measurement of:

- conformity,
- anchoring,
- dominance,
- authority effects,
- influence cascades,
- echo chambers,
- adversarial manipulation,
- false consensus.

5. Heterogeneous Multi-Model Evaluation

Study of interactions between agents powered by different LLM families and capability levels.

6. Multi-Architecture Evaluation

Evaluation across consultation, committee, adversarial and stakeholder-negotiation systems.

7. New Evaluation Metrics

Metrics focusing on the quality and causal structure of consensus formation, rather than only final task accuracy.

8. Open Research Framework

Potential release of SocietyXAI as a reusable Python framework and visualization environment.

32. What SocietyXAI Is NOT

SocietyXAI should not become:

«"Five LLMs talk and then vote."»

It is also not simply:

«"We visualize which agent spoke to which agent."»

And it should not claim causality merely because one message preceded another.

The central differentiator is:

«SocietyXAI actively intervenes on artificial societies and observes how alternative deliberation trajectories unfold.»

33. Final Research Vision

The ultimate goal is to treat a multi-agent generative system not merely as software architecture, but as an artificial society whose beliefs evolve through interaction.

Traditional evaluation asks:

«Did the agents reach the correct answer?»

SocietyXAI asks:

«Why did they reach that answer?»

And then goes further:

«Would they still have reached it if a particular agent, argument, evidence item, confidence signal, majority cue, or interaction order had been different?»

This transforms multi-agent explainability from passive observation into controlled causal investigation.

The broader research vision is therefore:

«SocietyXAI provides an experimental laboratory for studying how artificial societies form beliefs, influence one another, converge, fail, resist manipulation, and collectively reason.»

The most interesting outcome is not necessarily discovering that multi-agent systems reach consensus.

It is discovering what makes them change their minds.